

Chapter 6: CONFIDENCE INTERVAL

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Intervals estimation

Confidence intervals on the mean (expectation)

Case1: The variance σ^2 is known

Case2: The variance σ^2 is Unknown with large sample size $n \geq 30$

Case3: The variance σ^2 is Unknown but with small sample size $n < 30$

Confidence interval on population proportion

Confidence interval on the variance and standard deviation

Determining of the sample size

Determining of the confidence level

- We would like to examine a property X on a population.
- r.v. X follows a distribution $F(x; \theta)$, with an **unknown parameter θ** .
- A random sample of size n : $\mathbf{X} = (X_1, X_2, \dots, X_n)$.

Definition 1.1

An ***interval estimator*** of the parameter θ is a pair of statistics $L(X_1, \dots, X_n)$ and $U(X_1, \dots, X_n)$ of the random sample, satisfying

$$L(\mathbf{X}) \leq \theta \leq U(\mathbf{X}). \quad (1)$$

Then, with observed values $\mathbf{x} = (x_1, x_2, \dots, x_n)$,
 $[l(x), u(x)]$

is called an ***interval estimate*** of θ .

Definition 1.2 (Confidence interval - CI)

Consider a random sample $\mathbf{X} = (X_1, X_2, \dots, X_n)$ with the joint density probability function depending on a parameter $\theta \in \Theta$, and $L(\mathbf{X})$, $U(\mathbf{X})$ are two statistics such that $L(\mathbf{X}) \leq U(\mathbf{X})$. Then, a random interval

$$[L(\mathbf{X}), U(\mathbf{X})] \quad (2)$$

is called a **confidence interval for θ with confidence level $100 \cdot (1 - \alpha)\%$** ,
if

$$\mathbb{P}(L(\mathbf{X}) \leq \theta \leq U(\mathbf{X})) = 1 - \alpha. \quad (3)$$

Given an collection of observed values $\mathbf{x} = (x_1, x_2, \dots, x_n)$, we obtain an observed confidence interval for θ , that is $[l(\mathbf{x}), u(\mathbf{x})]$.

- ▶ We often choose the confidence level of $1 - \alpha = 0.95$ (0.9, or 0.99).

- ▶ **Interpreting of the confidence level $(1 - \alpha)$:**

Each time of sampling, we may obtain different observed values, thus, the corresponding calculated confidence intervals may differ from the others as well.

Nevertheless, within totally N times of sampling with sample size n , there are

- $(1 - \alpha) \cdot N$ times the true value of $\theta \in [l(\mathbf{x}), u(\mathbf{x})]$;
- $\alpha \cdot N$ times the true value of $\theta \notin [l(\mathbf{x}), u(\mathbf{x})]$;

Types of Confidence Interval

With a random sample $\mathbf{X} = (X_1, \dots, X_n)$, we are going to consider various types of confidence interval as follows:

1.) Confidence Intervals on the mean (expectation) μ :

- Case of known σ^2 (using the statistic Z);
- Case of UNKNOWN variance σ^2 with large sample size ($n \geq 30$) (using the statistic Z);
- Case of UNKNOWN variance σ^2 but with small sample size ($n < 30$) (using the statistic t -Student);

2.) Confidence Intervals on the proportion p ;

3.) Confidence Intervals on the variance σ^2 .

Confidence Intervals on the mean (expectation)

Problem

Consider a population X following a distribution with the unknown mean μ , and the variance σ^2 can be known or unknown.

Basing on a random sample $(X_1, X_2, \dots, X_n)$,
find a confidence interval of μ with given confidence level $(1 - \alpha)$.

Strategies

We consider this problem in three situations as follows

- ▷ Case1: The variance σ^2 is known.
- ▷ Case2: The variance σ^2 is Unknown with large sample size $n \geq 30$.
- ▷ Case3: The variance σ^2 is Unknown but with small sample size $n < 30$

Confidence intervals on the mean μ - theoretical background

Proposition 2.1

When the sample size is reasonably large $n \geq 30$ (or $n < 30$, X follows a normal distribution) and the variance σ^2 is known, the statistic

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

can be approximately calculated by the Gaussian distribution $\mathcal{N}(0, 1)$.

Definition 2.2

The (lower) quantile γ (with $\gamma \in (0, 1)$) of a random variable X is the value q_γ satisfying

$$\mathbb{P}(X \leq q_\gamma) = \gamma. \quad (4)$$

In case $X \sim \mathcal{N}(0, 1)$, we usually denote by z_γ for q_γ and look for the value z_γ in the table of the distribution of Φ .

(remind that Φ is the probability distribution function of Gaussian $\mathcal{N}(0; 1)$)

Some values of z_γ that are usually used:

γ	0.95	0.975	0.98	0.985	0.99	0.995
z_γ	1.64	1.96	2.05	2.17	2.33	2.58

Confidence intervals on the mean μ - theoretical background (cont.)

Construction of CI for μ

With **confidence level** $(1 - \alpha)$, we have

$$\mathbb{P}\left(-z_{1-\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha, \quad (5)$$

or

$$\mathbb{P}\left(\bar{X} - z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha, \quad (6)$$

where $z_{1-\frac{\alpha}{2}}$ is the quantile $1 - \frac{\alpha}{2}$ of the Gaussian distribution $\mathcal{N}(0, 1)$.

Confidence intervals on the mean μ - theoretical background (cont.)

- ▶ Given a random sample, a corresponding Confidence Interval for μ with confidence level $(1 - \alpha)$ is

$$\bar{X} - z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}. \quad (7)$$

- ▶ Given observed values, an estimate of Confidence Interval for μ with confidence level $(1 - \alpha)$ is

$$\bar{x} - z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}. \quad (8)$$

- ▶ Quantity $\varepsilon = z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$ is called estimation error of the confidence interval.

Confidence interval on the mean μ - theoretical background

In this case, we use the sample variance S^2 to replace σ^2 .

Remind that the sample variance S^2 is an estimate for $\text{Var}(X) = \sigma^2$.

Then, by the Central Limit Theorem, one gets the following proposition.

Proposition 2.3

*When the sample size is reasonably large $n \geq 30$
and the variance σ^2 is Unknown, the statistic*

$$\frac{\overline{X} - \mu}{S/\sqrt{n}},$$

can be approximately calculated by the Gaussian distribution $\mathcal{N}(0, 1)$.

Confidence interval on the mean μ - theoretical background (cont.)

Construction of CI on μ

With **confidence level** $(1 - \alpha)$, we have

$$\mathbb{P} \left(-z_{1-\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq z_{1-\frac{\alpha}{2}} \right) = 1 - \alpha, \quad (9)$$

or

$$\mathbb{P} \left(\bar{X} - z_{1-\frac{\alpha}{2}} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \cdot \frac{S}{\sqrt{n}} \right) = 1 - \alpha, \quad (10)$$

where $z_{1-\frac{\alpha}{2}}$ is the quantile $1 - \frac{\alpha}{2}$ of the Gaussian distribution $\mathcal{N}(0, 1)$.

Confidence interval on the mean μ - theoretical background (cont.)

- ▶ Given a random sample, a corresponding Confidence Interval for μ with **confidence level** $(1 - \alpha)$ is

$$\overline{X} - z_{1-\frac{\alpha}{2}} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{1-\frac{\alpha}{2}} \cdot \frac{S}{\sqrt{n}}. \quad (11)$$

- ▶ Given observed values, an estimate of Confidence Interval for μ with **confidence level** $(1 - \alpha)$ is

$$\overline{x} - z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \overline{x} + z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}.$$

- ▶ Quantity $\epsilon = z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$ is called **estimation error** of the confidence interval.

Confidence interval on the mean μ - theoretical background

In this case 3: the variance σ^2 is Unknown but with small sample size $n < 30$,
suppose that X follows a normal distribution, one has

Proposition 2.4

In this case, the statistic

$$T := \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim T(n-1),$$

where $T(n-1)$ is the t -Student distribution with $(n-1)$ degree of freedom.

Confidence interval on the mean μ - theoretical background (cont.)

Construction of CI for μ

With **confidence level** $(1 - \alpha)$, we have

$$\mathbb{P}\left(-t_{1-\frac{\alpha}{2}}^{n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{1-\frac{\alpha}{2}}^{n-1}\right) = 1 - \alpha, \quad (12)$$

or

$$\mathbb{P}\left(\bar{X} - t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{S}{\sqrt{n}}\right) = 1 - \alpha, \quad (13)$$

where $t_{1-\frac{\alpha}{2}}^{n-1}$ is the quantile $1 - \frac{\alpha}{2}$ of the ***t*-Student distribution** with $(n - 1)$ **degree of freedom**.

Confidence interval on the mean μ - theoretical background (cont.)

- ▶ Given a random sample, a corresponding Confidence Interval for μ with **confidence level** $(1 - \alpha)$ is

$$\bar{X} - t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{S}{\sqrt{n}}. \quad (14)$$

- ▶ Given observed values, an estimate of Confidence Interval for μ with **confidence level** $(1 - \alpha)$ is

$$\bar{x} - t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{s}{\sqrt{n}}.$$

- ▶ Quantity $\epsilon = t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{s}{\sqrt{n}}$ is called **estimation error** of the confidence interval.

Summary - Strategies to find a CI for mean μ

Step1. Compute the sample mean $\bar{x}$ and sample variance s^2 , from observed values $x_1, x_2, \dots, x_n$.

Step2. Determine which of the following 3 cases corresponds to:

Case 1 variance σ^2 is known, $n \geq 30$ (or $n < 30$, $X \sim \mathcal{N}(\mu; \sigma^2)$).

Case 2 Unknown variance σ^2 with large sample size $n \geq 30$.

Case 3 Unknown variance σ^2 but small sample size $n < 30$ and $X \sim \mathcal{N}(\mu; \sigma^2)$.

Step3. Find the quantile:

- $z_{1-\frac{\alpha}{2}}$ in **case1** or **case2**;
- $t_{1-\frac{\alpha}{2}}^{n-1}$ in **case3**.

Step4. Compute estimation error of CI

$$\epsilon = \begin{cases} z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}, & \text{in case1,} \\ z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}, & \text{in case2,} \\ t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{s}{\sqrt{n}}, & \text{in case3,} \end{cases}$$

Step5. $100(1 - \alpha)\%$ CI for the population mean is $\bar{x} - \epsilon \leq \mu \leq \bar{x} + \epsilon$.

Example 2.5

Suppose that the monthly salary of workers (million VND) in a factory follows a normal distribution. Choosing randomly 16 workers to make a survey and collect the following data

Monthly salary		0.8	1.0	1.2	1.3	1.5	1.7	2	2.3	2.5
Number of workers		1	1	2	2	2	3	2	2	1

- a.) Assume that $\sigma = 0.63$, find a 96% confidence interval for the mean of monthly salary of one worker in this factory.
- b.) Find a 99% for the mean of monthly salary.

Confidence interval on the mean μ - example 2.5 - a.)

Hint: a.)

Step1. Compute the sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k n_{i.} x_i = \frac{1}{16} \cdot (1 \times 0.8 + 1 \times 1.0 + \dots + 1 \times 2.5) = 1.625;$$

and the sample variance:

$$\begin{aligned} s^2 &= \frac{1}{n-1} \sum_{i=1}^k n_{i.} (x_i - \bar{x})^2 \\ &= \frac{1}{16-1} \left[1 \times (0.8 - 1.625)^2 + \dots + 1 \times (2.5 - 1.625)^2 \right] = 0.243. \end{aligned}$$

KConfidence interval on the mean μ - example 2.5 - a.)

Step2. As $n < 30$, X follows a normal distribution and the variance σ^2 is known (due to assumptions in question (a)), this corresponds to the **Case 1**.

Step3. The confidence level is $(1 - \alpha) = 0.96$, so $\alpha = 0.04$, thus one gets

$$z_{1-\frac{\alpha}{2}} = z_{0.98} = 2.05.$$

Step4. Compute the estimation error:

$$\epsilon = z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} = 2.05 \times \frac{0.63}{\sqrt{16}} = 0.323.$$

Step5. Conclusion: a 96% confidence interval for the mean of monthly salary of one worker is

$$[\bar{x} - \epsilon, \bar{x} + \epsilon] = [1.302, 1.948].$$

KConfidence interval on the mean μ - example 2.5 - b.)

Step2. Now, X follows a normal distribution, but the variance σ^2 is **unknown** and the sample size is small $n < 30$, so this corresponds to the case 3.

Step3. The confidence level $(1 - \alpha) = 0.99$, so $\alpha = 0.01$, thus one gets

$$t_{1-\frac{\alpha}{2}}^{n-1} = t_{0.995}^{15} = 2.95 \quad .$$

Step4. Compute the estimation error:

$$\epsilon = t_{1-\frac{\alpha}{2}}^{n-1} \cdot \frac{s}{\sqrt{n}} = 2.95 \times \frac{\sqrt{0.243}}{\sqrt{16}} = 0.364.$$

Step5. Conclusion: a 99% confidence interval for the mean of monthly salary of one worker is

$$[\bar{x} - \epsilon, \bar{x} + \epsilon] = [1.261, 1.989] \quad .$$

Confidence interval on population proportion p

Problem

Consider a population X , where the proportion of individuals possessing a property $\mathcal{A}$ is $p \in (0, 1)$.

Basing on a random sample $X_1, X_2, \dots, X_n$,

find a confidence interval for p with confidence level $(1 - \alpha)$.

Confidence interval on population proportion p - theoretical background

- ▶ Let Y be the number of individuals possessing the property $\mathcal{A}$ among n individuals of survey,
then $Y \sim \mathcal{B}(n, p)$.

- ▶ Let

$$\hat{P} = \frac{Y}{n}. \quad (15)$$

- ▶ The statistic $\hat{P}$ has the expectation and variance that are respectively

$$\mathbb{E}(\hat{P}) = \mu_{\hat{P}} = p, \quad \text{and} \quad \text{Var}(\hat{P}) = \sigma_{\hat{P}}^2 = \frac{p(1-p)}{n}.$$

Proposition 3.1

If $n.\hat{P} \geq 5$ and $n.(1 - \hat{P}) \geq 5$, then the statistic

$$Z = \frac{\hat{P} - \mu_{\hat{P}}}{\sigma_{\hat{P}}} = \frac{\hat{P} - p}{\sqrt{\frac{p.(1-p)}{n}}} \rightsquigarrow \mathcal{N}(0, 1), \quad (16)$$

and

$$W = \frac{\hat{P} - p}{\sqrt{\frac{\hat{P}.(1-\hat{P})}{n}}} \rightsquigarrow \mathcal{N}(0, 1). \quad (17)$$

Confidence interval on population proportion p - theoretical background (cont.)

Construction of CI for p

When the sample size n is reasonably large,
with **confidence level** $(1 - \alpha)$, we have

$$\mathbb{P} \left(-z_{1-\frac{\alpha}{2}} \leq \frac{\hat{P} - p}{\sqrt{\frac{\hat{P} \cdot (1 - \hat{P})}{n}}} \leq z_{1-\frac{\alpha}{2}} \right) \approx 1 - \alpha, \quad (18)$$

or

$$\mathbb{P} \left(\hat{P} - z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{P} \cdot (1 - \hat{P})}{n}} \leq p \leq \hat{P} + z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{P} \cdot (1 - \hat{P})}{n}} \right) = 1 - \alpha, \quad (19)$$

where $z_{1-\frac{\alpha}{2}}$ is the quantile $1 - \frac{\alpha}{2}$ of the **Gaussian distribution** $\mathcal{N}(0, 1)$.

Confidence interval on population proportion p - theoretical background (cont.)

- Given a random sample, a corresponding confidence interval for p with **confidence level** $(1 - \alpha)$ is

$$\hat{P} - z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{P} \cdot (1 - \hat{P})}{n}} \leq P \leq \hat{P} + z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{P} \cdot (1 - \hat{P})}{n}}. \quad (20)$$

- Given observed values, an estimate of Confidence Interval for p with **confidence level** $(1 - \alpha)$ is

$$\hat{p} - z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}} \leq p \leq \hat{p} + z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}.$$

- Quantity $\epsilon = z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}$ is called **estimation error** of the confidence interval.

Confidence interval on population proportion p - Strategies

B1. Compute the sample proportion: $\hat{p}$.

B2. Verifying the conditions: $n \cdot \hat{p} \geq 5$ and $n \cdot (1 - \hat{p}) \geq 5$.

B3. Find the quantile $z_{1 - \frac{\alpha}{2}}$ in the table of Gaussian distribution.

B4. Compute the estimation error:

$$\epsilon = z_{1 - \frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}.$$

B5. Conclusion: a $100 \cdot (1 - \alpha)\%$ confidence interval for the population proportion is:

$$\hat{p} - \epsilon \leq p \leq \hat{p} + \epsilon.$$

Example 3.2

Suppose that the monthly salary of workers (in million VND) in a factory follows a normal distribution. Choosing randomly 16 workers to make a survey and collect the following data

Monthly salary		0.8	1.0	1.2	1.3	1.5	1.7	2	2.3	2.5
Number of workers		1	1	2	2	2	3	2	2	1

A work is considered to have a high salary if his or her monthly salary is at least 2 million VND. Find a 95% confidence interval for the proportion of workers having high salaries.

Hint: From observed values, compute the sample proportion $\hat{p} = \frac{5}{16}$, then, verifying the conditions $n \cdot \hat{p} = 5$ and $n \cdot (1 - \hat{p}) = 9$.

Example 3.3

In a forest, guardians track a bird species by putting an electronic collar on them. They have put the collars on 1000 individuals. After that, capturing 200 individuals and found that 40 of them wearing the collars. Find a confidence interval for the proportion of individuals wearing the collars in the forest with confidence level 95%.

Example 3.4

In a factory, at the product quality examination stage, 100 products in a shipment are randomly chosen, and 20 poor quality products are detected.

- 1) Find a 95% confidence interval for the proportion of poor quality products in a shipment.
- 2) With confidence level 99%, if one would like to have an estimation error as 0,04, how many products need to be examined?

Confidence interval on the variance and standard deviation

Problem

Consider a population X following a normal distribution with unknown variance σ^2 .

Basing on a random sample $(X_1, X_2, \dots, X_n)$, find a confidence interval for the variance σ^2 and also for the standard deviation σ with given confidence level $(1 - \alpha)$.

Remind that for a normally distributed sample, the probability distribution of the sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$, is as follows

Proposition 4.1

For a random sample $X_1, X_2, \dots, X_n$ chosen from a normal distribution $\mathcal{N}(\mu, \sigma^2)$,

$$\frac{(n-1) \cdot S^2}{\sigma^2} \sim \chi^2(n-1).$$

Confidence interval on the variance and s.d. - theoretical background

Construction of CI for σ^2

With **confidence level** $(1 - \alpha)$, we have

$$\mathbb{P} \left(\chi_{\frac{\alpha}{2}}^2 (n-1) \leq \frac{(n-1).S^2}{\sigma^2} \leq \chi_{1-\frac{\alpha}{2}}^2 (n-1) \right) = 1 - \alpha, \quad (21)$$

or

$$\mathbb{P} \left(\frac{(n-1).S^2}{\chi_{1-\frac{\alpha}{2}}^2 (n-1)} \leq \sigma^2 \leq \frac{(n-1).S^2}{\chi_{\frac{\alpha}{2}}^2 (n-1)} \right) = 1 - \alpha, \quad (22)$$

and

$$\mathbb{P} \left(\sqrt{\frac{(n-1).S^2}{\chi_{1-\frac{\alpha}{2}}^2 (n-1)}} \leq \sigma \leq \sqrt{\frac{(n-1).S^2}{\chi_{\frac{\alpha}{2}}^2 (n-1)}} \right) = 1 - \alpha, \quad (23)$$

where $\chi_{\gamma}^2 (n-1)$ is the quantile γ of the **Chi-squared distribution with** $(n-1)$ **degree of freedom.**

Confidence interval on the variance and s.d. - theoretical background (cont.)

- Given a random sample, a corresponding Confidence interval for σ^2 with **confidence level** $(1 - \alpha)$ is

$$\frac{(n-1).S^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)} \leq \sigma^2 \leq \frac{(n-1).S^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}. \quad (24)$$

- Given observed values, an estimate of Confidence Interval for μ with **confidence level** $(1 - \alpha)$ is

$$\frac{(n-1).s^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)} \leq \sigma^2 \leq \frac{(n-1).s^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}. \quad (25)$$

Corollary: Confidence interval for the standard deviation σ

- Given a random sample, a corresponding Confidence interval for σ with confidence level $100.(1 - \alpha)\%$ is

$$\sqrt{\frac{(n-1).S^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)}} \leq \sigma \leq \sqrt{\frac{(n-1).S^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}}. \quad (26)$$

- Given observed values, an estimate of Confidence Interval for σ with confidence level $100.(1 - \alpha)\%$ is

$$\sqrt{\frac{(n-1).s^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)}} \leq \sigma \leq \sqrt{\frac{(n-1).s^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}}. \quad (27)$$

Example 4.2

In candy factory, an automatic process is programmed to package candy bags weighing 52gr, with an acceptable standard deviation of ± 1 g. An examiner has doubts about the operation of the packaging machine, and as a result some packaged candy bags may have weighed much less or heavier than supposed. For verification of these doubts, the examiner chose randomly 10 packaged candy bags in a shipment, and compute the sample variance as 4,2gr.

Find a 95% confidence interval for the standard deviation and make a conclusion that whether this packaging machine works well or not?

■ Motivation:

Before sampling, for a given confidence level, the estimate confidence interval is expected to have a length that should not get over a specific value.

Then, a raising question: what the smallest sample size should be taken?

■ Problem:

Find the smallest value of sample size n so that $\epsilon \leq \epsilon_0$, with given ϵ_0 and α .

Determining of the sample size

- a. **In case $\text{Var}(X) = \sigma^2$ is known**, from the formula

$$\epsilon = z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}},$$

to get $\epsilon \leq \epsilon_0$, we need to choose n satisfying

$$n \geq (z_{1-\frac{\alpha}{2}})^2 \cdot \frac{\sigma^2}{\epsilon_0^2}.$$

- b. **In case $\text{Var}(X) = \sigma^2$ is unknown**, we use the sample variance s^2 as an estimate for σ^2 . Then, the sample size n is chosen to satisfy:

$$n \geq (z_{1-\frac{\alpha}{2}})^2 \cdot \frac{s^2}{\epsilon_0^2}.$$

Determining of the sample size

a. In case $\hat{p}$ is known, to get $\epsilon \leq \epsilon_0$, from the formula

$$\epsilon = z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}} \Rightarrow n \geq \left(z_{1-\frac{\alpha}{2}}\right)^2 \cdot \frac{\hat{p} \cdot (1 - \hat{p})}{\epsilon_0^2}.$$

b. In case $\hat{p}$ is unknown, we have $\epsilon = z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}$.

Since $\hat{p} \cdot (1 - \hat{p})$ has the maximum as 0.25, with $\hat{p} = 0.5$, thus

$$\epsilon \leq z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{0.25}{n}}.$$

Hence, to get $\epsilon \leq \epsilon_0$, we need to choose n so that $z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{0.25}{n}} \leq \epsilon_0$, i.e.

$$n \geq \frac{0.25 \cdot \left(z_{1-\frac{\alpha}{2}}\right)^2}{\epsilon_0^2}.$$

Determining of the confidence level

■ Motivation:

Now, fixing the sample size n , we would like to construct a confidence interval whose length consists with a given specific value.

Then, a raising question: **what is the value of confidence level?**

■ Problem:

Find $(1 - \alpha)$, with given n and ϵ .

Determining of the confidence level

- a. **In case $\text{Var}(X) = \sigma^2$** is known, from the formula

$$\epsilon = z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}},$$

we get

$$z_{1-\frac{\alpha}{2}} = \frac{\epsilon \cdot \sqrt{n}}{\sigma}.$$

Then, using $z_{1-\frac{\alpha}{2}} \Rightarrow$ obtain the confidence level $(1 - \alpha)$
(using the probability distribution function of Gaussian $\mathcal{N}(0; 1)$).

- b. **In case $\text{Var}(X) = \sigma^2$ is unknown**, we compute the sample standard deviation s . From that, determine $z_{1-\frac{\alpha}{2}}$ by

$$z_{1-\frac{\alpha}{2}} = \frac{\epsilon \cdot \sqrt{n}}{s}.$$

Then, using $z_{1-\frac{\alpha}{2}} \Rightarrow$ obtain the confidence level $(1 - \alpha)$.
(using the probability distribution function of Gaussian $\mathcal{N}(0; 1)$)

Determining of the confidence level

Starting from the formula

$$\epsilon = z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\widehat{p} \cdot (1 - \widehat{p})}{n}},$$

we obtain

$$z_{1-\frac{\alpha}{2}} = \epsilon \cdot \sqrt{\frac{n}{\widehat{p} \cdot (1 - \widehat{p})}}.$$

Then, using $z_{1-\frac{\alpha}{2}} \Rightarrow$ obtain the confidence level $(1 - \alpha)$.
(using the probability distribution function of Gaussian $\mathcal{N}(0; 1)$)

Example 6.1

Monitoring 1000 electronic equipment, found that 825 equipment failed within 10 years of use.

- a.) Find a 95% confidence interval for the proportion of failed equipment.
- b.) To get the estimation error less than 0.03, how many equipment need to be monitored at least with in 10 years?

Hint: Let $p \in (0, 1)$ be the proportion of equipment failed within 10 years of use. Then, the sample proportion is $\hat{p} = \frac{825}{1000} = \frac{33}{40}$.

Since $n \cdot \hat{p} \geq 5$ and $n \cdot (1 - \hat{p}) \geq 5$, we get that $\frac{\hat{P} - p}{\sqrt{\frac{\hat{P} \cdot (1 - \hat{P})}{n}}}$ can be approximately calculated by the normal distribution $\mathcal{N}(0, 1)$.

Exercise 1

After a harvest, weighing the fruits and get the following data

X (g)	200 - 210	210 - 220	220 - 230	230 - 240	240 - 250
Number of fruits	12	17	20	18	15

- (a) Compute the sample mean and the sample standard deviation.
- (b) Find a 99% confidence interval for the mean of the fruits' weight.
- (c) With confidence level 99%, to get an estimation error not exceed 2gr, at least how many fruits that we need to monitor?

Exercises (cont.)

Exercise 2

Collect the height X (in centimeter) of teenagers in a resident area, and obtain

X (cm)	140-145	145-150	150-155	155-160	160-165	165-170
Nb of teens	1	3	7	9	5	2

- (a) Use an appropriate type of graphs to show that the data is sampled from a normal distribution.
- (b) Compute sample parameters.
- (c) Estimate the mean of height of teenagers in this area with confidence level 99%.
To get an estimation error as 1 (cm), we need to collect data on how many teenagers' heights?
- (d) Teenagers with heights of at least 160 (cm) are classified as health class A. Find a 95% confidence interval for the proportion of teenagers in health class A.

Exercise 3

A new medicine is used as treatment for 50 patients of disease B, resulting in 40 patients curing this disease.

- a) Estimate the proportion of patients curing the disease with confidence level 95% and 99%, respectively.
- b) To get an estimation error not exceed 0,02 with confidence level 95%, how many patients need to be surveyed at least?

Exercise 4

The proportion of titanium in an alloy used in the casting of aerospace parts is measured in 51 of a randomly selected sample with a sample standard deviation $s = 0,37$. Find a 95% confidence interval for σ .